HEBEI HANDAN DAMINGCHANG BIOGAS RECOVERY AND UTILIZATION PROJECT

Document Prepared by China Classification Society Certification Co. Ltd.

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Project Title	**Hebei Handan Damingchang Biogas Recovery and Utilization Project**
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Report Title	**Validation Report for Hebei Handan Damingchang Biogas Recovery and Utilization Project**

Client	Climate Bridge (Shanghai) Ltd.
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Summary:

- A brief description of the validation and the project

China Classification Society Certification Co. Ltd. (CCSC), commissioned by Climate Bridge (Shanghai) Ltd. has performed the validation of Hebei Handan Damingchang Biogas Recovery and Utilization Project on the basis of requirements of Verified Carbon Standard (VCS) Version 4.4.

The Project, which is located in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China, is to utilize animal manure waste from Daming farms as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas will be used for electricity generation. The project involves the construction and operation of the anaerobic animal manure management system, the biogas pre-treatment system, the organic fertilizer production plant, the biogas power generation system and Emergency flare system. The intended total installed capacity is 2.33 MW (1.165MW×2). An average annual emission reduction of 33,017 tCO_2e and a total emission reduction of 231,119 tCO_2e during the first 7-year crediting period will be achieved, and 16,510 MWh of electricity will be supplied to the North China Power Grid (NCPG) each year. The project will displace the equivalent power generated by the existing power plants and likely capacity additions within the given grid and avoids GHG emissions of methane through recovery and destruction of biogas.

- The purpose and scope of validation

The validation is an independent third party assessment of the project's baseline, estimated GHG emission reductions or net anthropogenic GHG removals, the monitoring plan and the crediting period using the applicable VCS requirements. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated Verified Carbon Units (VCUs).

The scope of the validation is defined as an independent and objective review of the Project Description (hereafter refer to as "PD"), the project's baseline study and monitoring plan, the emission reduction calculation spreadsheet and other relevant documents refer to VCS Standard Version 4.4 and all the GHG program requirements. In order to confirm that the project activity, as documented, is rational and meets the identified criteria. The validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented.

Validation is not meant to provide any consultancy towards the project participants. It is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project is valid to be submitted for VCS registration.

- The method and criteria used for validation

Validation is conducted using China Classification Society Certification Co. Ltd. (CCSC) procedures in line with the requirements specified in the latest version of the VCS Validation Manual and applying auditing techniques. The validation team assessed the project activity's compliance against the VCS Standard Version 4.4, the selected CDM methodology and the project description. The validation criteria followed the guidance documents provided by VCS including the VCS Standard Version 4.4, VCS Program Guide Version 4.3 and the applied CDM methodologies **AMS-III.D: Methane recovery in animal manure management systems (Version 21.0) and AMS-I.D:Grid connected renewable electricity generation (Version 18.0)**.

- The number of findings raised during validation

In the course of the validation, 2 Corrective Action Requests (CARs), 1 Clarification Request (CL) were raised and successfully closed. The assessment is included in the report.

- Any uncertainties associated with the validation

There are no restrictions of uncertainty for validation.

- Summary of the validation conclusion

"Hebei Handan Damingchang Biogas Recovery and Utilization Project" (VCS ID 3969) with regard to the relevant requirements of VCS standard Version 4.4. CCSC confirms all validation activities including objectives, scope and criteria, reasonableness of assumptions, project description, monitoring plan adhere to VCS Standard Version 4.4 and all associated updated as documented in this report, are complete. In conclusion, it is CCSC's opinion that the project "**Hebei Handan Damingchang Biogas Recovery and Utilization Project**" as described in the PD version 2.0 dated 17-February-2024, meets all relevant requirements for VCS activities and all relevant host Party criteria and correctly applies the methodologies **AMS-III.D (Version 21.0) and AMS-I.D (Version 18.0)**.

CONTENTS

1 INTRODUCTION

1.1 Objective

Climate Bridge (Shanghai) Ltd. has commissioned the CCSC to performed the validation of Hebei Handan Damingchang Biogas Recovery and Utilization Project (hereafter referred to as "the Project") on the basis of requirements of Verified Carbon Standard (VCS) Version 4.4/5/.

CCSC as the validation body (VB) of the project has been accredited as a DOE by UNFCCC and meets the competence requirements as set out in ISO 14065:2020.

The objective of the Validation is to have an independent evaluation of a project activity by a designated operational entity against the requirements of the VCS/5//12//13//14/ and GHG program applied, on the basis of the project design document. In particular, the project's baseline, monitoring plan, and the project's compliance with relevant VCS requirements, GHG program requirements and host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Verified Carbon Units (VCUs).

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the project description (PD) to validate that (a) the project design is actual, (b) the baseline scenario is correctly defined as per the applied methodology and related tools, (c) the project is additional, (d) the monitoring plan can be implemented and is transparent and adequate and (e) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature. The PD is reviewed against the criteria stated in VCS standard Version 4.4/5/and the approved baseline and monitoring methodology AMS-III.D (Version 21.0) and AMS-I.D (Version 18.0)/6//7/. The validation was based on the requirements VCS Validation and Verification Manual and VCS Standard version 4.4 applying auditing techniques. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

1.3 Reasonableness of Assumptions

The estimated GHG emission reductions and removals of the project activity are calculated on the basis of assumed scenarios as per the applied methodology. CCSC has undertaken a reasonable assessment of the applicability of the methodology, the baseline scenario selection, the values and sources of the parameters used for calculating GHG emission reductions from the project activity, in accordance with VCS Standard (version 4.4) /5/ and all the GHG program requirements, to assure the assumptions have been justified correctly. The detailed validation is described in this report, section 3.3. .

1.4 Summary Description of the Project

The project is implemented by Daming Jiujin Energy Co., Ltd, in order to collect animal manure of dairy cows from Daming farms in Daming County and use animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas will be used for electricity generation at the project site. The electricity generated by the project will be delivered to North China Power Grid (NCPG). The residual waste from the CSTR digesters will be used to produce organic fertilizer at the project site. The project started construction on 06-February-2022 and was put into operation on 26-August-2022, on which the project began generating GSG emission reductions or removals and was considered as the Project start date as per the VCS standard V4.4/5/.

Prior to the implementation of the project, the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

The Project is expected to avoid GHG emission of methane through recovery and destruction of biogas. In addition, the electricity generated from the biogas will be delivered to NCPG to replace equivalent amount of electricity generated by the fossil fuel fired power plants connected to the grid. The project is expected to achieve an annual average emission reduction of 33,017 tCO_2e and a total emission reduction of 231,119 tCO_2e during the first 7-year crediting period.

2 VALIDATION PROCESS

2.1 Method and Criteria

The overall validation, from Contract Review to Validation Report and opinion, was conducted using CCSC internal procedures. The process includes three phases:

1) Document review;

2) On-site visit and follow-up interviews;

3) Issuance of the final Validation Report and opinion.

After commissioned by Climate Bridge (Shanghai) Ltd., a validation team was appointed by CCSC. A project specific validation plan, which includes the site visit schedule and the list of evidence to be gathered, was then developed to guide the validation auditing process to ensure efficiency and effectiveness. The site visit was conducted on 20-March-2023, which was described in detail in section 2.3 of this report. During the site visit, a desk review of critical evidences, an interview with key personnel and a thorough project site inspection were carried out by the assessment team. The validation opinion was then derived from the independent and objective assessment on the documents and information gathered from PP as per the VCS requirements.

The purpose of the validation is to present a risk assessment for determining the nature and extent of validation procedures necessary to ensure the risk of auditing error is reduced to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. Therefore, validation of the project description and verification of the monitoring plan and the reported project results were measured for compliance against the criteria including VCS standard version4.4/5/, VCS registration and issuance process version 4.3/13/, VCS program guide version 4.3/12/, VCS project description template version 4.2/35/ and other relevant CVS requirements.

2.2 Document Review

The PD version 2.0/3/ dated 17-February-2024 and the PD version 1.0, version 1.1/1/ dated 08-November-2022 & 25-March-2023, in particular the applicability of the methodology, the baseline determination, the additionality analysis, the project start date, the monitoring plan, and the emission reduction estimation provided in the form of a spreadsheet/2//4, were assessed as part of the validation based on the latest version of VCS requirements. All the information described in the PD/3/ was reviewed and cross-checked with evidence available to the validation team.

Appendix B of this report contains a complete list of all documents and proofs reviewed by the validation team.

2.3 Interviews

On 20-March-2023, CCSC performed an on-site visit at the physical site of the project.

During the on-site visit, the team has interviewed with key personnel from the project owner, local environmental protection bureau officer, local Stakeholders, and the VCU buyer. The assessment content and topics/the persons interviewed during the on-site inspection are provided in the table below .

Date: 20-March-2023	
Interview topics	**Interviewed Organizations and persons**
--The status of VCS project. --The evidences of implementation of key equipment,	**Daming Jiujin Energy Co., Ltd (Project Owner)**

Date: 20-March-2023	
Interview topics	**Interviewed Organizations and persons**
--Main equipment and technical parameters --Project boundary. --Monitoring plan, monitoring device installation and calibration --Data processing activities, --Monitoring data collection. --Quality Management; organizational structure, responsibilities and competencies; Internal QA/QC Management procedures and document control.	Li Jihui, General manager Li Rui, Operating staff
--Compliance with National Laws and Regulations. --Conditions prior to project construction. --Status of landfill site. --Operation of landfill site. --The process and participation of the stakeholder consultation. --The impact of the project activity. --The complaint by local stakeholders and the implementation of the mitigation measures.	**Local environmental protection bureau:** Guo Yulun, **Local residents:** Cao Suqiao Yang Zhirong
-- Applicability of selected methodology -- Baseline of the project -- Emission factors -- Preparation of PD --Compliance of the monitoring plan with the monitoring methodology --Compliance of data monitoring with the monitoring plan --Assessment of data and calculation of GHG emission reductions	Climate Bridge (Shanghai) Ltd. (Buyer) Ding Wenjun, Project manager

2.4 Site Visits

The validation site visit was conducted on 20-March-2023 at project site in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China. A ground inspection of the project was carried out during the site visit. VVB visited project sites and inspected the implemented facilities as described in section 3.3 of PD version1.1 dated 25-

March-2023. VVB interviewed the project implementer, operation staffs, local officers,local Stakeholders and VCUs buyer. The interviewed personnel and objective are listed in the table in section 2.3 of this report. During the site inspection, the project site was inspected and documents evidence were checked. Details and inspection procedures are summarized in the following table:

No.	Activity performed on-site	Site location	Date
1	Opening meeting: introduce the objective and the scope of the on-site inspection	Office in Daming Jiujin Energy Co., Ltd.	20-March-2023
2	Interview with key personnel of PP, local government officer, local stakeholders, VCUs buyers	Office in Daming Jiujin Energy Co., Ltd.	20-March-2023
3	Project site inspection, check the facilities on site, interview with technical expert and operating staff	Project site, Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China	20-March-2023
4	Document check	Project site, Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China	20-March-2023
5	Finding summary	Office in Daming Jiujin Energy Co., Ltd..	20-March-2023
6	Close meeting	Office in Daming Jiujin Energy Co., Ltd.	20-March-2023

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

A corrective action request (CAR) is raised if one of the following occurs:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

The validation team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for registration.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for CCSC' s positive conclusion on the project design.

The Corrective Action Requests and Clarification Requests raised by CCSC were resolved during communications between the project owner and CCSC to guarantee the transparency of the validation. The concerns raised, and the responses from PP are summarized in the table of appendix C.

The final PD Version 2.0 /3/serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

There are 2 CARs and 1 CLs raised during the validation. The assessment is included in the report. The details of the findings are listed in the appendix C at the end of this report.

2.5.1 Forward Action Requests

No Forward Action Requests were raised during the validation

3 VALIDATION FINDINGS

3.1 Project Details

- Project type, technologies and measures implemented, and eligibility of the project

Through the onsite visit, interviewing the project developer and reviewing the PD/1//3/, it was confirmed that the project is to collect animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas will be used for electricity generation. Therefore, it was confirmed that the sectoral scope is scope 01: Energy industries (renewable - / non renewable sources) and scope 13: Waste handling and disposal as per the UNFCCC Standard “Applicability of sectoral scopes version 01.0”.

It was confirmed that the project is not a grouped project.

The information and descriptions reported in section 1.1 of the VCS PD/1//3/ have been checked. The project involves the construction and operation of a centralized animal manure management system which collects manure waste (dairy cow) from Daming farms located in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China . The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The digestate from the anaerobic digesters will be sent to digestate advanced treatment units, it mainly involved filter press and then return to nearby field. The anaerobic digesters are expected to produce 8,760,000 m^3 biogas per year according to the project approval/17/. Before electricity generation, the biogas will be processed through Biogas pre-treatment system to remove impurities and moisture etc., in order to prevent the corrosion of the project facility. And the digestate from the anaerobic digesters, both solid and liquid digestate, will be used as material to produce organic fertilizer at the onsite organic fertilizer production plant.

The project also involves the construction and operation of a biogas power plant. The recovered biogas from pre-treatment system is then used for electricity generation at the biogas power plant. The intended installed capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), The biogas power plant of the project is expected to supply 16,510 MWh of electricity to the North China Power Grid annually.

In addition, a emergency flare system is installed to ensure all the biogas will be destroyed in case the generator is defected. The power rate is 2 kW, and the design discharge is 1000m^3/h.

The technology is confirmed by checking the facilities during on-site inspection and cross-checked with the FSR/24/, Contract and Annex for Construction of Anaerobic System, Sales contract of container-type biogas cogeneration unit, and the Technical Agreement for container-type biogas cogeneration unit Procurement /19/.

The project is expected to achieve an average annual emission reduction of 33,017 tCO_2e and a total emission reduction of 231,119 tCO_2e during the first 7-year crediting period.

The PP has described and justified how the project is eligible under the scope of the VCS Program in section 1.3 of the PD/1//3/ as per the section 2.1.1 of VCS standard Version 4.4. The assessment is provided as below,

1. The project activity generates GHG emission reductions or removals including CO_2 and CH_4, which belong to the seven Kyoto Protocol greenhouse gases. The applied

methodology AMS-III.D (Version 21.0) and AMS-I.D (Version 18.0) of the project are methodologies approved under CDM Program, which is a VCS approved GHG program

2. “The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.” Via checking the PD/1//3/ of the project, it is verified that the project is implemented to utilize animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas for electricity generation, the digestate from the anaerobic digesters will be sent to digestate advanced treatment units, it mainly involved filter press and then return to nearby field., and the additionality is verified as actual, thus there is no assumption of having generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction for this project.

3. “The VCS Program also excludes the following project activities under the circumstances indicated in Table 1” of section 2.1.3 of VCS Standard Version 4.4, via checking all the excluded project activities in the table, it is verified that this project type is not excluded by the scope of VCS program.

In conclusion, the project is eligible to the scope of the VCS Program as per the VCS Standard Version 4.4.

- Project design, including eligibility criteria for grouped projects

The project has been designed to be a single installation of an activity, not multiple project activity instances or as a grouped project.

- Project proponent and other entities involved in the project

Through document review and site interview, the validation team confirmed the details of the project proponent as:

Organization name	Daming Jiujin Energy Co., Ltd
Role in the project	Project proponent
Contact person	Zhang Kai
Title	General Manager
Address	Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China
Telephone	+86 21 23019950

Email	3542346576@qq.com

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	VCUs Buyer
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86 21 6246 2036
Email	gao.zhiwen@climatebridge.com

- Ownership

Via checking the Environmental Impact Assessment (EIA) report and approval/16//23/, Project approval/17/, and the business license of Daming Jiujin Energy Co., Ltd./15/, it is verified that the PP has the legal right to control and operate the project activity. By checking the Project Construction Permits /18/ and equipment purchase contract /19/, the validation team confirm that the project proponent has the legal right to construct the project. Furthermore, via checking the application for State Grid connection approved by NCPG/20/ and the Power Purchase Contract/21/ signed between Daming Jiujin Energy Co., Ltd and NCPG, CCSC confirmed that Daming Jiujin Energy Co., Ltd. has the legal right to operate the project activity.

In conclusion, the validation team confirm the project ownership of Daming Jiujin Energy Co., Ltd and its legal right to control and operate the Project.

- Project start date

The project start date is 26-August-2022 which is verified as the date on which the project began generating GHG emission reductions or removals as per the concept in VCS Standard Version 4.4.

Via checking Application for grid connection approved by NCPG/20/ and the Acceptance report of the test operation for the generator set/22/, it is verified that 26-August-2022 is the date when all the system started fully operation and started gird-connected power generation, hence the start date is correct and in line with the VCS standard.

- Project crediting period

This project adopts renewable crediting periods of 7 years from 26-August-2022 to 25-August-2029 (both days included).

- Project scale and estimated GHG emission reductions or removals

The annual GHG emission reduction of the project is estimated to be 33,017 tCO_2e which is less than 300,000 tonnes of CO_2e per year. Therefore, the validation team confirms that the project scale falls in the category of Projects.

Project Scale	
Project	✓
Large project	

Through checking emission reductions calculation spreadsheet provided by PP/4/, the validation team confirms that the estimated GHG Emission Reductions of the project is as follows:

Year	Estimated GHG emission reductions or removals (tCO_2e)
26-August-2022~31-December-2022	11,579
01-January-2023~31-December -2023	33,017
01-January -2024~31-December -2024	33,017
01-January -2025~31-December -2025	33,017
01-January -2026~31-December -2026	33,017
01-January -2027~31-December -2027	33,017
01-January -2028~31-December -2028	33,017
01-January -2029~25-August -2029	21,438
Total estimated ERs	231,119
Total number of crediting years	7
Average annual ERs	33,017

- Project location

Through satellite maps and on-site visit, the project is located in Cao Ren Village, Dangtou Township, Daming County, Handan City, Hebei Province, P.R China. The geographical coordinates for the project site are east longitude 115º17’0.405’’ and north latitude 36º11’20.403’’. the geographical coordinates for the involved farm (Daming farm) are 115°

16'56.52"E and 36° 11'32.56"N, the depth of baseline lagoon is 2.05m and has already replaced by the project. The project and farm location was confirmed during site visit by GPS devices.

- Conditions prior to project initiation

Via on-site interview with villagers, local government officer and checking the Environmental Impact Assessment (EIA) /23/ and the FSR/24/, CCSC confirmed that the scenario prior to the implementation of the proposed project activity is that electricity supplied from the NCPG connected fossil fuel fired power plants and the animal manure waste was left to decay in anaerobic manure management system (anaerobic lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility.

The baseline scenario is the same as the scenario existing prior to the implementation of the project activity, which has been confirmed during the site interview with local officer and checking the Environmental Impact Assessment (EIA)/23/ and the FSR/24/. Thus, it is verified that the baseline scenario is reasonable and correct.

Therefore, the project is confirmed not to be implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

- Project compliance with applicable laws, statutes and other regulatory frameworks

The PD/1//3/ has stated that the project is in the field of renewable energy, this industry is confirmed as in line with the laws and regulations as listed in the PD/1//3/.

Also, the project is legally approved by local government by checking the project approval/17/, Approval for Grid-connection/20/ and Environmental Impact Assessment (EIA) approval /16/, which means that the project is in line with the national and local laws and regulations based on the local expertise of the VVB. By interviewing with local officer, CCSC verified that the project will be subject to regular inspection by local government during the implementation and operation period to ensure continuous compliance, which is consistent with the statement in the PD /1//3/.

In conclusion, the project complies with all relevant local, regional and national laws, statutes and regulatory frameworks in China.

- Participation under other GHG programs:
    - Projects registered (or seeking registration) under other GHG program(s)

    The validation team checked the registry of CDM EB (http://cdm.unfccc.int), Gold Standard (http://www.goldstandard.org), China Certified Emission Reduction (CCER) Program (http://cdm.ccchina.org.cn/ccer.aspx) and other GHG program. It is confirmed that the project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program.

    - Rejection by other GHG programs

    The project has neither been rejected by any other GHG programs which has been confirmed via checking the UNFCCC, GS, VCS and other GHG scheme website.

- Other forms of credit and supply chain (Scope 3) emissions":

  - Emissions trading programs and other binding limits

  The validation team confirms that the Net GHG emission reductions generated by the project will not be used for compliance with an emissions trading program or to meet binding limits. The validation team checked Chinese Emission Trading System (China ETS) and found that the project is not accredited/registered under China ETS. Thus, the project activity is not involved in other Emissions trading programs or other binding limits. Therefore, it is confirmed that the emission reductions will not be double counted.

  - Other forms of environmental credit sought or received and eligible to be sought or received

  The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

  Renewable energy certificates are available for trading in the host country. However, the same is not availed by the PP. The undertaking regarding the same is submitted by PP which is acceptable to the validation team. The validation team also checked the REC website and found the declaration to be correct.

  - Supply chain (Scope 3) emissions

  According to the Verra Releases Updates to VCS Program updated on 21-December-2022 and VCS standard version 4.4, issuance of public statement(s) to help prevent Scope 3 emissions double claiming and Email notification of the potential risk of Scope 3 emissions double claiming will be effective from 01-July-2023 onwards. Thus, this section is not required.

Via interview with the project proponent and based on above assessment, it is confirmed that the project has not sought or received any other form of environmental credit.

- Sustainable development contributions

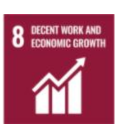
Via on-site inspection, checking the evidence provided and interviewing with the project implementer, CCSC confirmed that the project contributes the sustainable development through the following aspects:

- Ensure access to affordable, reliable, sustainable and modern energy for all (Goal 7 of UN SDG,)–verified by checking the approval for Grid-connection/20/, the power purchase contract/21/ and the operation report of the biogas power generator/22/.

- This project will increase income of local residences and accelerate economy development in rural areas. During the crediting period, direct and indirect employment opportunities will be generated. Thus, the project will achieve SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all". Via interview with staffs during on-site inspection, the assessment team verified that the project create full employment positions regardless of gender or other status. Hence the contribution to SDG 8 is confirmed.

- The project will achieve a GHG emission reduction during the crediting period. Thus, the project will achieve SDG 13“Take urgent action to combat climate change and its impacts”. By checking the FSR/24/, the ER calculation spreadsheet/4/, and on-site inspection, it is verified that a total of 231,119 tCO_2e will be achieved in a 7-year crediting period. Hence the contribution to SDG 13 is confirmed.

The details of the project contribution to sustainable development are listed in the following table:

SDG	Indicators	Chinese Sustainable Development Progress	Project activity contribution
	SDG 7 “Ensure access to affordable, reliable, sustainable and modern energy for all”	By the end of 2015, China had fully solved the problem of providing electricity to all people without electricity, and reached the goal of the Sustainable Development Agenda 15 years ahead of schedule. From 2015 to 2020, the share of non-fossil energy increased from 14.5 % to 19.6%, while the share of raw coal decreased from 72.2 % to 67.6%%.	The project is to capture and utilize biogas for power generation and accordingly substitute equivalent electricity from thermal power. This contributes to achieve China’s stated sustainable development priorities “By 2030, the proportion of non-fossil energy consumption will reach about 25%”.
	SDG 8 “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”[1].	China continuously improves the quality and efficiency of development. In-depth implementation of the innovation-driven development strategy, the rapid development of small and medium-sized enterprises. Adhering to	The project activity will increase employment opportunities for the operation of the project activity. This contributes to one of China’s actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent

[1] https://sdgs.un.org/goals/goal8

		the policy of giving priority to employment, the unemployment rate has remained at a low level. By coordinating epidemic prevention and control and economic and social development, it has become the only major economy to achieve positive growth in 2020 and has made positive contributions to the recovery of the global economy.	work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".
	SDG 13 "Take urgent action to combat climate change and its impacts"[2].	In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP was reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule.	The project uses CSTR to treat animal manure, collect and destroy the generated biogas to avoid methane emissions. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas"

- Additional information relevant to the project, including:
  - Leakage management for AFOLU projects

  It is not applicable as the project is not a AFOLU project,

  - Commercially sensitive information

  No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the validation team for analysis which lead to positive conclusion for this validation

[2] https://sdgs.un.org/goals/goal13

Based on above assessment and demonstration, it is concluded that the description in the PD version 2.0 dated 17-February-2024/3/ is accurate, complete, and the nature of the project is actual and understandable. And it is confirmed that the project has been implemented as described in the project description via site inspection.

2 CARs were raised and successfully closed. Refer to Appendix C for detailed assessment.

3.2 Safeguards

3.2.1 No Net Harm

The Environmental Impact Assessment (EIA) Report/23/ of the project activity was compiled by Hebei Qizheng Environmental Technology Co., Ltd. and approved by Administrative Examination and Approval Bureau of Daming County on 09-September-2021, with approval No. "Dashenhuanping [2021] No.18"/16/. The assessment team confirms all environmental impacts has been fully analysed and no net harm was detected. Refer to section 3.3.3 of this report for detailed environmental impacts arising from the Project construction and operation phase. The qualification certificate of the EIA report preparer was also checked to ensure the authority/25/.

Through interview the PP, the governmental officer and local residents, it is confirmed by the assessment team that the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 3.1 of this report above.

In conclusion, it is confirmed by the assessment team that the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to hold the relevant stakeholder's consultation prior to the project construction. The validation team checked the relevant dates during the desk review and site inspection. A stakeholder survey was carried out by the project developer on 08-July-2021 which was prior to the start date of construction indicated in the Commencement Report of project /26/.

The project owner invited local stakeholders to participate in the questionnaire survey by posting announcements on the bulletin boards of the village, phone calls and SMS. The survey was conducted through distributing and recollecting responses of filled questionnaires. The questionnaire was reasonably designed to assess the project impacts on the local environment and social economic development. In total, 30 questionnaires were distributed and recollected with a 100% response rate.

By checking the filled survey questionnaires/27/, it was verified that all respondents knew or heard about the Project. Most of them are supportive of the project construction, and agree that the Project will bring more benefit than loss. And the survey shows that many local stakeholders think the Project will help improve the life of local people and promote local economic development without any adverse environmental impact. Therefore, the validation team confirm that the local stakeholder has no negative comments for the construction of the project activity. Via interview with the local officer and local residents during site visit, and checking the responsive questionnaires, it is verified that there were no negative comments received during the implementation period. Furthermore, it is also confirmed, through interviewing with the PP during site visit, that the PP will set up an on-going communication mechanism to hold stakeholder meetings every year to issue questionnaires and to answer the questions from various stakeholders.

In conclusion, CCSC confirms that the local stakeholder consultation has been conducted adequately and appropriately as per VCS standard/5/. And no need to change the project design based on the stakeholder inputs.

1 CL was raised and successfully closed. Refer to Appendix C for detailed assessment.

3.2.3 Environmental Impact

In China, an Environmental Impact Assessment approval or an Environmental Registration is required for a new construction project, according to Chinese legislation. The validation team reviewed the EIA Report/23/ and EIA Approval/16/ for Project, and confirms the correctness of the approaches used by PP against the adverse environmental impacts. The EIA report was prepared by Hebei Qizheng Environmental Technology Co., Ltd. Thus the validation team checked the certificate of the report preparer and confirmed Hebei Qizheng Environmental Technology Co., Ltd. is qualified to carry out the environmental impact assessment/25/. The assessment results showed that the environmental impact caused by the project will be minor. In conclusion, the PP have followed the requirements of the host country with regards to addressing environmental impacts as below:

1. <u>Construction Phase</u>

1.1 Air pollution

Construction dust and road dust during the construction has a certain effect to the surrounding areas of construction site, these adverse effects are accidental, temporary and partial, as long as to take effective measures and strengthen management, the scope of its influence is generally limited to the surrounding area of the construction site and will disappear along with the end of construction.

1.2 Wastewater

Wastewater during the construction period mainly includes construction wastewater and domestic sewage. The following measures are taken to prevent and control the pollution of construction wastewater: construct temporary diversion ditches on the construction site; Set up a sedimentation tank, and reuse the washing water for construction machinery as much as possible after simple treatment of equipment and vehicles;

Domestic sewage can be collected and processed by setting up mobile toilets, and regularly handed over to the sanitation department to remove garbage and excrement, which can effectively prevent the sewage generated by construction workers from polluting the water environment.

1.3 Noise

The noise generated by the project construction has a slight impact on the surrounding sensitive points, and its pollution impact is localized and short-term. After adopting a reasonable construction organization method, its impact on the surrounding area can be reduced to acceptable range.

1.4 Solid waste

During the construction period, a certain amount of construction waste and domestic waste from construction workers will be generated, of which part of the construction waste will be recycled, and the unusable waste will be handed over to the sanitation department for disposal. After the above measures are taken, the solid waste of the project will not cause pollution to the surrounding environment.

2. Operation Phase

2.1 Air pollution

The air pollution during the operation mainly includes the exhaust gas of the biogas generator and the flare, and dust from organic fertilizer plant. The exhaust gas is to be treated through 2 stage purification equipment before being discharged through exhaust cylinders with 20m high. The dust from organic fertilizer plant is to be filtered by dust collector and reused in the organic fertilizer plant. The NO_x, sulfur dioxide, H_2S, NH_3 and particulate matter in all the exhaust gas by the project meets Grade 2 emission standard of "Integrated Emission Standard of Air Pollutants " (GB16297).

2.2 Wastewater

The wastewater during operation is mainly domestic sewage and biogas slurry from anaerobic digesters. The domestic sewage of the project is pretreated in a septic tank and reused in anaerobic digesters. Most biogas slurry will also be recycled in anaerobic digesters, surplus biogas slurry will be used as material for organic fertilizer production. No wastewater will be discharged to the environment.

2.3 Noise

The noise of this project comes from noise from various mechanical operation and vibration. All production equipment is placed in the workshop or steel container, and measures are taken to reduce vibration and noise. The operating noise is effectively attenuated after being blocked by the solid wall.

2.4 Solid waste

The solid waste of the project is mainly domestic garbage, filtered dust. The domestic garbage and filtered dust are collected uniformly and then sent to landfill site for landfill treatment.

Hazardous waste: The hazardous waste, e.g. mineral oil, generated by the project mechanical facilities shall be properly collected and stored in the hazardous waste temporary deposit, and shall be submitted to the relevant qualified entities for centralized treatment regularly.

Via checking the EIA report and EIA approval, interviewing the local environmental government officer and local residents, and on-site inspection, the validation team confirmed that appropriate measures stated in PD, which is complies with the EIA report, have been taken to eliminate or minimize the environmental impact caused by the Project.

In conclusion, the environmental impact during the project construction will be temporary and not significant, and the environmental impact during the project operation will be minor. The project activity can reduce greenhouse gas emissions and environmental pollution caused by methane release and coal-fired power generation. The Project owner takes appropriate actions to minimize adverse environmental impacts.

3.2.4 Public Comments

The project sought the registration with the VCS Program. According to VCS requirements, the Project Description was submitted for public comment period. The project was allocated with VCS ID 3969. The public comment period started on 13-December-2022 and ended on 12-January-2023 (https://registry.verra.org/app/projectDetail/VCS/3969). No comments were received during this period which has been verified by checking the dedicated website.

3.2.5 AFOLU-Specific Safeguards

This section is not required as the project is not AFOLU project

3.3 Application of Methodology

3.3.1 Title and Reference

The assessment team checked that following methodology and tools are applicable for the project activity as below:

AMS-III.D Methane recovery in animal manure management systems (version 21.0)/6/;

AMS-I. D Grid connected renewable electricity generation (version 18.0)/7/;

Project and leakage emissions from anaerobic digesters (version 02.0)/9/;

Tool to calculate the emission factor for an electricity system (version 07.0)/8/;

Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0)/10/

3.3.2 Applicability

Demonstration and justification regarding the applicability of the methodology and tools selected by the project proponent are shown in the table below.

Table 3-1 Assessment of Applicability conditions of AMS-III.D (version21.0).

No.	Applicability conditions	Justifications	Validation Opinion
1	This methodology covers project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant	Applicable. The project is a centralized plant treats animal manure waste collected from Damingchang Farms. The project replaces existing anaerobic animal manure agement systems (open lagoon) in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane.	By checking the FSR/24/ and site visit, it is able to confirm: the project involves the construction and operation of the anaerobic animal manure management system, the biogas pre-treatment system, the organic fertilizer production plant, the emergency flare system and the biogas power generation system. Therefore, this criterion is applicable.
2	This methodology is only applicable under the following conditions: a) The livestock population in the farm is managed under confined conditions	Applicable. All the livestock population in the farms within the project boundary is managed under confined conditions	By checking the FSR, the dynamic table of Daming farm/31/ and on-site inspection, it is able to confirm that the livestock population in the farm is managed under confined conditions Therefore, this criterion is applicable.
	b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied	Applicable. As per the FSR, manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries)	By checking the FSR, interviewing with the officer from local Environmental Bureau and on-site inspection, it is able to confirm: Manure or the streams obtained after treatment are not discharged into natural water resources Therefore, this criterion is applicable.
	c) The annual average temperature of baseline site where anaerobic manure treatment	Applicable. The annual average temperature of baseline	By checking the FSR/24/, and interviewing the local environmental officer during

	facility is located is higher than 5°C	site where anaerobic manure treatment facility is located is higher than 5°C. The annual average temperature at the project site is 13.5℃.	site visit, it is able to confirm: The annual average temperature at the project site is 13.5℃ Therefore, this criterion is applicable.
	d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m	Applicable. In the baseline scenario the retention time of manure waste in the anaerobic lagoons is greater than one month, and their depths are at least 1 m.	By interviewing with operating staff of the project during site inspection, it is confirmed that in the baseline scenario, the retention time of manure waste in the anaerobic lagoons is greater than one month, and the depth of the baseline lagoon is 2.05m. Therefore, this criterion is applicable.
	e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	Applicable. No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.	By checking the FSR and interview with villagers during site visit, it is able to confirm that there is no methane recovery and destruction by flaring or combustion in the baseline scenario Therefore, this criterion is applicable.
3	The project activity shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured;	Applicable. The residual waste from the animal manure management system of the project will be used to produce organic fertilizer, which is handled aerobically and will not result in methane emissions.	By checking the FSR and on-site inspection, it is able to confirm the organic fertilizer production plant has been implemented for producing organic fertilizer, which is handled aerobically and does not result in methane emissions. Therefore, this criterion is applicable.

	(b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	Applicable. Biogas produced by the project are used directly to produce electricity, the emergency flare ensure biogas would be destroyed while exigencies happened	By checking the FSR, the Purchase Contract and Acceptance Report of the Anaerobic System/19//32/ and on-site facility inspection, it is able to confirm the biogas produced by the digester is used for electricity generation or flared in case of exigencies. Therefore, this criterion is applicable.
	(c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply.	Applicable. The storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester.	By checking the FSR, the EIA of the project and interview with operation staff during site visit, it is able to confirm that the storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester. Therefore, this criterion is applicable.
4	Projects that recover methane from landfills shall use “AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use “AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology “AMS-III.AO Methane recovery through controlled anaerobic digestion".	Irrelevant. The project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters.	By checking the FSR and site visit, it is able to confirm that the project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters. Therefore, this criterion is irrelevant.
5	Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the	Applicable. Part of the recovered biogas is used to produce and deliver electricity to the power grid, therefore	By checking the FSR/24/, the power purchase contract/21/ and on-site inspection, it is able to confirm that part of the recovered biogas is used to produce and deliver electricity

	recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	AMS.I. D is applied. Part of the biogas is used to power auxiliary equipment of the project activity, zero is used as its emission factor, and this part of biogas is not eligible as an SSC CDM Type I project component, which means no emission reductions will be claimed for this part of biogas utilization.	to the State Grid, and part of the biogas is used to power auxiliary equipment of the project activity, hence the emission factor for the 2nd part of biogas utilization is zero and no emission reductions will be claimed for it. Therefore, this criterion is applicable.
6	New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".	Applicable. The project is a Greenfield project. The emission reduction sourced from methane recovery is 33,017 tCO_2e/yr, which is lower than the threshold of 60,000 tCO_2e/yr; the total installed capacity of the project is 2.33MW, which is lower than the threshold of 15MW. Therefore, the project is in line with "General Guidelines to SSC CDM methodologies"	By checking the ER calculation spreadsheet, the FSR and facilities on-site, it is able to confirm that the project complies with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".
7	The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	The project is a Greenfield project, and does not involve the replaced equipment; therefore, this is irrelevant.	By checking the FSR and site visit, it is able to confirm that no equipment needs to be replaced. Therefore, this criterion is irrelevant.
8	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.	Applicable. The emission reductions from the recovery and destruction of methane (viz. Type III components of the project) are estimated as 33,017	By checking the ER calculation spreadsheet and site visit, it is able to confirm that the emission reductions from the recovery and destruction of methane (viz. Type III components of the

		tCO_2e/yr, which is less than 60 kt CO_2 equivalent.	project) are 33,017 tCO_2e/yr, which is less than 60 kt CO_2 equivalent. Hence, this criterion is applicable.
Conclusion: Methodology AMS-III.D (version21.0). is applicable to the project activity.			

Table 3-2 Assessment of Applicability conditions of AMS-I.D (version18.0)

No.	Applicability conditions	Justifications	Validation Opinion
1	This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	Applicable. The project will use biogas to generate electricity, and the electricity will be supplied to North China Power Grid.	By checking the FSR/24/, approval for Grid-connection/20/, the power purchase contract/21/ and site visit, it is able to confirm the project uses biogas to generate electricity, and the electricity is supplied to NCPG. Hence, this criterion is applicable.
2	Illustration of respective situations under which each of the methodology (i.e., “AMS-I.D.: Grid connected renewable electricity generation”, “AMS-I.F.: Renewable electricity generation for captive use and mini-grid” and “AMS-I.A.: Electricity generation by the user) applies is included in the appendix.	The illustration of respective situations for AMS-I.D stated in the appendix is the same as applicability condition No. 1 above. No need for further justification.	same as No.1 in this table
3	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	Applicable. The project is a greenfield plant.	By checking the FSR and site visit, it is able to confirm the project is a greenfield plant. Hence, this criterion is applicable.

4	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m2.	Irrelevant. The project is not a hydro power plant.	By checking the FSR and on-site inspection, it is able to confirm the project is not a hydro power plant.
5	If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve non-renewable components.	By checking the FSR and site visit, it is able to confirm the project does not involve non-renewable components.
6	Combined heat and power (co-generation) systems are not eligible under this category.	Applicable. According to the FSR, the only output for the project is electricity and the project does not involve co-generation system.	By checking the FSR and on-site inspection, it is able to confirm the project does not involve co-generation system. Hence, this criterion is applicable.
7	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct1 from the existing units.	Irrelevant. The project does not involve capacity addition.	By checking the FSR and site visit, it is able to confirm the project does not involve capacity addition.

8	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve retrofit, rehabilitation or replacement.	By checking the FSR and site visit, it is able to confirm the project does not involve retrofit, rehabilitation or replacement.
9	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	Irrelevant. The project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects.	By checking the FSR and site visit, it is able to confirm the project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects.
10	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Irrelevant. The project does not involve biomass.	By checking the FSR and site visit, it is able to confirm the project does not involve biomass.
Conclusion: Methodology AMS-I.D (version18.0) is applicable to the project activity.			

In addition, the project meets the applicability conditions of the applied tools as follows:

	Tool to calculate the emission factor for an electricity system (version 07.0)		
No.	**Applicability conditions**	**Justifications**	**Validation Opinion**

1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects)	Applicable. The project utilizes biogas for power generation to supply electricity to NCPG that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project.	By checking the FSR, the Acceptance report of test operation of generator set/22/,, the operation log/33/ and site visit, it is able to confirm that the project utilizes biogas for power generation to supply electricity to NCPG that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project.
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	In the host country as off-grid power generation is not significant. Therefore, emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied.	According to the situation in the host country and based on the VVB’s expertise, the emission factor for the project electricity system can be calculated only for grid power plants.
3	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable. The project electricity system is located in a non-Annex I country.	By checking the Annex I country and site visit, it is able to confirm the project is not located in an Annex I country.

4	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project does not involve biofuels consumption.	By checking the FSR and interview with operation staff during site visit, it is able to confirm that the project does not involve biofuels consumption.
	Project and leakage emissions from anaerobic digesters (version 02.0)		
1	The following sources of project emissions are accounted for in this tool: (a) CO_2 emissions from consumption of electricity associated with the operation of the anaerobic digester; (b) CO_2 emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH_4 emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH_4 emissions from flaring of biogas.	Applicable. All sources of project emissions have been accounted.	By checking the ER calculate sheet and the PD/3/, it is able to confirm that all sources of project emissions have been accounted.
2	The following sources of leakage emissions are accounted for in this tool: (a) CH_4 and N_2O emission from composting of digestate; (b) CH_4 emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.	Irrelevant. The project does not involve composting, anaerobic decay or anaerobic storage of digestate.	By checking the FSR and site visit, it is able to confirm that the project does not involve composting, anaerobic decay or anaerobic storage of digestate.
3	Emission sources associated with N_2O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.	Applicable. As per the applied methodology, N_2O emissions are neglected because these are minor emission sources.	By checking the FSR, and interview with operation staff during site visit, it is able to confirm the N_2O emissions are neglected because these are minor emission sources.

	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)		
1	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	Applicable. The source of electricity consumption of the project is Scenario A: Electricity consumption from the grid	By site visit, it is able to confirm that scenario A: electricity consumption from the grid is applicable to the source of electricity consumption of the project.
2	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the	Applicable. The electricity generated by the project is supplied to the NCPG (Scenario I).	By checking the approval of grid-connection, the power purchase contract /20//21/ and site visit, it is able to confirm the electricity generated by the project is supplied to the NCPG, which is in

	grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.		line with Scenario I.
3	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	Applicable. No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.	By on-site inspection, it is able to confirm that no captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.

In conclusion, the validation team hereby confirms that the selected methodologies and tools are approved CDM methodologies and tools, and are applicable to the project activity.

3.3.3 Project Boundary

The project boundary basically defines the physical and geographical boundary of the project facility and it is well defined in the PD version 2.0/3/ section 3.3 according to methodologies AMS-III.D. and AMS-I.D.The project boundary includes:

(a) Physical and geographical sites of the livestock;

(b) Centralized animal manure management system;

(c) The biogas power plant and Emergency flare system;

(d) All the power plants physically connected into the NCPG

Via site inspection and checking the FSR/24/, it is verified that project boundary is clearly defined in the PD version2.0 /3/ as per the applicable methodologies /6//7/., and emission sources included in the project boundary have been appropriately justified in the PD version2.0 as well /3/.

The main emission sources and gases that included or excluded in the project boundary are listed in the table below:

Source		Gas	Included?	Justification/Explanation
Baseline	Direct emissions from the manure	CO_2	No	Excluded for simplification.
		CH_4	Yes	The major source of emissions in the baseline

Source		Gas	Included?	Justification/Explanation
	treatment processes	N_2O	No	Excluded for simplification.
	Emissions from electricity consumption /Generation	CO_2	Yes	The major source of emissions in the baseline
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
Project	Physical leakage of biogas in the manure management systems	CO_2	No	Excluded for simplification.
		CH_4	Yes	Main emission source.
		N_2O	No	Excluded for simplification.
	Emissions from flaring or combustion of the gas stream	CO_2	No	Excluded for simplification.
		CH_4	No	Excluded for simplification. The enclosed flare is installed for emergency, the emissions are assumed to be very small. The project will not claim this part of emission reduction
		N_2O	No	Excluded for simplification.
	Emissions from use of fossil fuels or electricity	CO_2	Yes	The project consumes electricity during operation, so emission from use of electricity is the main emission source. The project does not involve fossil fuel consumption, so the emission from use of fossil fuels is not included.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
	Emissions from incremental transportation distances	CO_2	No	Excluded for simplification.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
	Emissions from the storage of	CO_2	No	Excluded for simplification.
		CH_4	No	The storage time of the manure after removal

Source		Gas	Included?	Justification/Explanation
	manure			from the animal barns, including transportation, is within 24 hours before being fed into the anaerobic digester. hence Emissions from the storage of manure is not accounted for.
		N_2O	No	Excluded for simplification.

The validation team confirms that GHG sources included and excluded from the project boundary is defined as correct and corresponding to the actual status of the project, through on-site inspection and checking the FSR.

By checking EIA of the project and operation log of collecting manure and sending the manure to digester, the information above has been validated, Thus, there will be no PEstorage related emissions. Meanwhile, the validation team has visited the relevant personnel and confirmed that the manure was transported into the system through a closed transportation pipeline within 24 hours, normally 12 hours, but not exceed 24 hours. and there are no related emissions during the entire project's raw material collection and other processes. It is verified that all the relevant sources have been selected and justified.

Therefore, CCSC concluded that the project boundary, the GHG and the corresponding sources are correctly justified for the project.

3.3.4 Baseline Scenario

As per para. 17 of AMS-III.D, the baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere.

As per para. 19 of AMS-I. D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

In the PD version 2.0/3/, it is stated that the baseline scenario of the Project is the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility; and equivalent amount of electricity was supplied from the NCPG connected with fossil fuel fired power plants. .

Via checking FSR, document review and on-site visit, it is confirmed that the baseline scenario is the same as the scenario prior to the Project construction, which is realistic and in line with the approved methodologies and tools applied for the Project. The validation team also confirms that the Project has been approved by Chinese government by checking the Project approval/17/ and Environmental Impact Assessment (EIA) approval/16/. It is concluded that the project is in complicate with all laws and regulations in China.

In conclusion, CCSC confirms the correctness and appropriateness of the baseline scenario, as proposed in the PD/3/, for the project activity.

3.3.5 Additionality

As per para. 15-16 of applied methodology AMS-III.D, project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the “Guidelines on the demonstration of additionality of small-scale project activities”. This additionality condition also applies to greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.

Via checking FSR, technical agreement, equipment purchase contract, and facility inspection during site visit/19//21/, it is confirmed that the total installed capacity of the power generation system is 2.33 MW, which is less than 5MW. The regulations relative to the project in China are identified as below.

I) Law of the People's Republic of China on Environment Protection;

II) Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution;

III) Law of the People's Republic of China on the Prevention and Control of Atmospheric Pollution;

IV) Regulations on prevention and control of pollution from large scale livestock and poultry breeding;

V) Discharge standard of pollutants for livestock and poultry breeding (GB 18596-2001);

VI) Technical standard of pollution prevention for livestock and poultry breeding (HJ/T 81-2001).

The project is to build a centralized anaerobic animal manure treatment system which collects manure waste (dairy cow) from Daming farms in Daming County, Handan CIty, Hebei province, P.R.China. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. Digestate from the anaerobic digesters will be sent to advanced treatment unit, in mainly involved the filter press as soon as digestate is removed from the digesters, The recovered biogas is used for electricity generation at the project site. Prior to the implementation of the project, the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants. By interviewing the project owner and employees, it is confirmed by the assessment team that the project collects manure waste from only Daming Farms in Daming district to recover biogas and generate electricity. Digestate from the anaerobic digesters will be sent to advanced treatment unit, Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. By checking Project Approval, FSR, and Technical agreement/16//24/, the total intended installed capacity of the project activity is 2.33 MW(2*1.165MW), which is less than 5 MW. Therefore, the project is deemed automatically additional.

Furthermore, by checking applicable regulations, the assessment team confirms that there are no laws or regulations that require the collection and destruction of methane from livestock manure. Thus, the project activity is not mandatory by any law, statute or other regulatory framework. Therefore, it is concluded that the project has fulfilled the VCS requirements regarding regulatory surplus.

3.3.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation and confirmed that the evaluation of baseline emissions, project emissions and leakage is as per the approved methodology and formulae used for calculation are correct. The detail analysis is as below:

1. Baseline Emissions

As per the methodology AMS-III.D and AMS-I. D, baseline emissions are composed of Methane emissions from the manure treatment processes in the absence of the project activity and Emissions associated with electricity generation using fossils or supplied by the Grid in the absence of the project activity. The baseline emissions are determined according to equation (1) :

$$BE_y = BE_{CH4,y} + BE_{Elec,y} \tag{1}$$

Where:

BE_y	=	Baseline emissions in year *y* (t CO_2e/yr)
$BE_{CH4,y}$	=	Baseline emissions from the manure treatment processes in year *y* (t CO_2e/yr)
$BE_{Elec,y}$	=	Baseline emissions associated with electricity generation in year *y* (t CO_2/yr)

Via reviewing the PD verision2.0/3/, it is confirmed the ERs is calculated as per the applied methodologies (AMS-III.D ver. 21.0 and AMS-I. D ver.18.0) with following steps listed below.

1.1 Baseline emissions from the manure treatment processes in year y ($BE_{CH4,y}$)

According to the applied methodology AMS-III.D (ver. 21.0), there are two options for the baseline emissions calculation:

(a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (B_o);

(b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content.

Option (a) is applied to calculate the baseline emissions from the manure treatment processes using formula (2) shown below:

$$\mathrm{BE}_{\mathrm{CH4,y}} = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j} \qquad (2)$$

Where:

GWP_{CH4}	=	Global Warming Potential (GWP) of CH_4 applicable to the crediting period (tCO_2e/tCH_4)
D_{CH4}	=	CH_4 density (0. 67kg/m^3 at room temperature (20ºC) and 1 atm pressure)
$\mathrm{UF_b}$	=	Model correction factor to account for model uncertainties (0.94)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
$\mathrm{MCF_j}$	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j
$\mathrm{B_{0,LT}}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m^3CH_4/kg-dm)
$\mathrm{N_{LT,y}}$	=	Annual average number of animals of type LT in year y (numbers)
$\mathrm{VS_{LT,y}}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$\mathrm{MS\%_{Bl,j}}$	=	Fraction of manure handled in baseline animal manure management system j

-producing capacity of the manure ($B_{o,LT}$) varies by species and diet. The preferred method to obtain $B_{o,LT}$ measurement values is to use data from country-specific published sources, measured with a standardised method ($B_{o,LT}$ shall be based on total as-excreted VS). These values shall be compared to IPCC default values and any significant differences shall be explained. If country specific B_o values are not available, default values from tables 10 of 2019 Refinement to the 2006 IPCC Guidelines volume 4 Chapter 10 can be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site.

b) VS are the organic material in livestock manure and consist of both biodegradable and non-biodegradable fractions. For the calculations, the total VS excreted by each animal species is required.

According to the AMS-III.D ver. 21.0, if country specific VS values are not available, IPCC default values from 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10 table 10.13A can be used provided that the project participants assess the suitability of those data to the specific situation of the treatment site particularly with reference to feed intake levels. The Chinese specific VS values are not available and then IPCC default values should be used. As for this Project, the genetic source of dairy cow is unknown and identified as Asian, it is conservative and the average mass of dairy cow at the project site is 360kg, Thus, VS value from 2019 Refinement to the 2006 IPCC Guidelines Volume 4 chapter 10 table 10.13A.default value can be used.

c) $B_{o,LT}$ values or VS values applicable to developed countries can be used provided the following four conditions are satisfied:

i. The genetic source of the livestock originates from an Annex I Party;

ii. The farm uses formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;

iii. The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);

iv. The project specific animal weights are more similar to developed country IPCC default values.

The Chinese specific B_0 values are not available. Hence IPCC default values should be used. The genetic source of dairy cow is unknown and identified as Asian and lower productivity, it is conservative. Thus, the value of 0.13 m^3CH_4/kg-VS for B_o can be used in ER estimation.

The animal weight is 360kg, according to equation 10.22A from 2019 Refinement to the 2006 IPCC Guidelines, VS can be calculated as follows

$VS_{LT,y}= VS_{rate} * TAM/1000 * 365$

VS_{rate} is 9 kg VS (1000kg animal mass)/day for dairy cow from Asia, which is default value from 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10 table 10.13A

TAM is animal mass and 360kg for the project dairy cow.

Thus, $VS_{LT,y}$ = 9kgVS(1000kg animal mass)/day/*360kg/1000 *365 days = 1,182.6kgVS/animal/year

d) Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which B_o is achieved. Where available country-specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10 can be used. The site annual average temperature is taken from official data at the nearest meteorological station, or from data available from historical on site observations. The project site is locates in warm temperate dry region. Thus, the MCF is 76%.

e) The annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365}) \quad (3)$$

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

1.2 Baseline emissions associated with electricity generation in year y (BEelec,y)

As per AMS-I.D, baseline emissions associated with electricity generation in year y is calculated according to equation (3) as below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y} \quad (4)$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

Calculation of combined margin CO_2 emission factor for grid ($EF_{grid,y}$)

According to "Tool to calculate the emission factor for an electricity system" (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project baseline emissions:

- **Step 1:** Identify the relevant electricity systems;
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- **Step 3:** Select a method to determine the operating margin (OM);
- **Step 4:** Calculate the operating margin emission factor according to the selected method;
- **Step 5:** Calculate the build margin (BM) emission factor;
- **Step 6:** Calculate the combined margin (CM) emission factor.

As China DNA has published the calculation method for emission factor of grid, the published data and method have been applied for this project to calculate operating margin (OM) and build margin, as following steps:

Step 1: Identify the relevant electricity systems;

This project site is in Hebei Province of China, which belongs to North China Power Grid (NCPG) according to the public delineation of DNA[3], so NCPG is identified as the relevant electric system.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

For this project, Option I (only grid power plants are included in the calculation) is chosen.

Step 3: Select a method to determine the operating margin (OM);

Calculation of Operating Margin should be based on one of the four following methods according to the tool:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch Data Analysis OM, or

(d) Average OM.

[3] http://cdm.ccchina.gov.cn/zyDetail.aspx?newsId=46143&TId=161

As the low-cost / must-run resources constituted less than 50% of total electricity generation of NCPG in recent five years (respectively 10.10%, 8.94%, 7,21%,6,39% and 5,92% in 2017, 2016, 2015, 2014 and 2013)[4], the Simple OM (a) method is selected and the following data vintage is chosen to calculate the emission factor:

Ex ante option: use a 3-year generation-weighted average, based on the most recent data available, without requirement to monitor and recalculate the emissions factor during the crediting period. And according to the tool, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

Step 4: Calculate the operating margin emission factor according to the selected method;

The Simple OM emission factor ($EF_{OM,y}$) is calculated as the generation-weighted average emissions per electricity unit (tCO_2e/MWh) of all generating sources serving in the system, excluding low-operating cost and must-run power plants. It may be calculated:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power plant / unit, or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

(a) The necessary data for Option A is not available; and

(b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(c) Off-grid power plants are not included in the calculation.

In this project, all of the above conditions can be met, so Option B was chosen.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO_2,i,y})}{EG_y}$$

(5)

Where:

[4] China Electric Power Yearbook (2014, 2015, 2016, 2017, 2018)

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO_2 emission factor in year y (t CO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type i in year y (t CO_2/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in **Step 3**.

Based on the most recent three years (2015-2017) where the data are the latest and available at the time of this PD submission, the calculation result of $EF_{grid,OM,,y}$ is 0.9419 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China[5].

Step 5: Calculate the build margin (BM) emission factor;

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including

[5] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In line with China's Baseline emission factors of regional grids 2019 (BEF_{2019}) published by the Development and Reform Commission of China, Option 1 is chosen for the project; the BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET \geq 20\%}$, in MWh);

(c) From $SET_{5\text{-}units}$ and $SET_{\geq 20\%}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample\text{-}CDM}$) the annual electricity generation ($AEG_{SET\text{-}sample\text{-}CDM}$, in MWh); If the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample\text{-}CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM->10yrs}$).

The BM emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} \quad (6)$$

where:

$EF_{grid,BM,y}$	=	Build margin emission factor of NCPG (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	=	Power units included in the build margin
y	=	The most recent year for which the generation data is available

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07 (07.0) is adopted following the clarifications[6] given by the CDM EB concerning the BM emission factor calculation:

(1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.

(2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.

[6] "Request for clarification on use of approved methodology AM0005 for several projects in China", the EB's guidance on DNV deviation request. http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQKK7WYJ

(3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into "grouped new power plant" according to their construction year, their province and their fuel type. The annual net electricity generation in the year y of each "grouped new power plant" $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, as the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (7)$$

where:

$EG_{m,y}$	=	Annual net electricity generation the unit m in year y (MWh)
CAP_m	=	Installed capacity of the unit m (MW)
$H_{m,y}$	=	Utilization hour of the unit m in the year y (h), determined according to the average utilization hour of the same type of unit in the same province
y	=	The most recent year for which the generation data is available. For the calculation of BM in 2019, y = 2017
m	=	grouped new power plant

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into "a grouped new power plant", CAP_m represents the total installed capacity of fuel type k power plants located in the provinces A and in the year t:

$$CAP_m = CAP_{A,t,k} \quad (8)$$

where:

CAP_m	=	Installed capacity of the unit m (MW), with m representing the specified combination of A, t, and k
$CAP_{A,t,k}$	=	Total installed capacity of fuel type k power plants located in the province A and in the year t
A	=	Provinces covered by the NCPG, namely, Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province and Inner Mongolia Autonomous Region
t	=	Years related to the grouped new power plants, for the 2019

calculation, t represents 2017, 2016, 2015…. Until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of the NCPG

k = Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others.

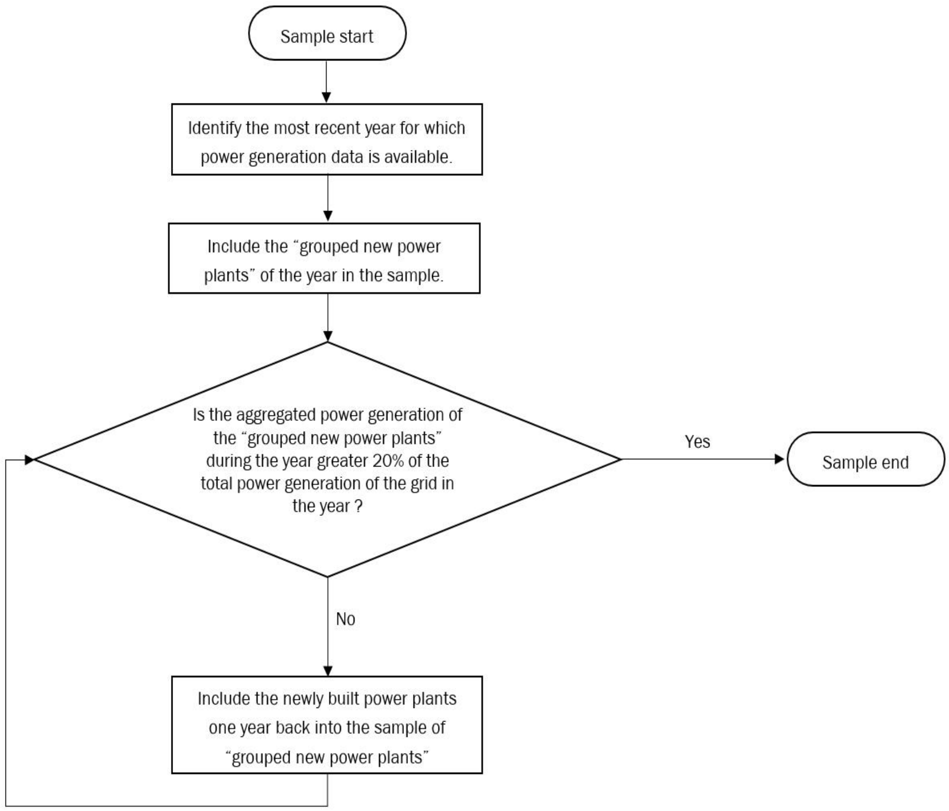


Figure 4 Procedure to determine the sample group of power units m used for the BM emission factor calculation

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to the Option A2 in the TOOL07, as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \qquad (9)$$

where:

$EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y (t CO_2/MWh)

$EF_{CO2,m,i,y}$ = Average CO_2 emission factor of fuel type i used in power unit m in year y (t CO_2/GJ)

$\eta_{m,y}$ = Average net energy conversion efficiency of power unit m in year y (ratio)

m = All power units serving the grid in year y except low-cost / must-run power units

3.6 Conversion factor (GJ/MWh)

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation takes the following form due to conservativeness:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \quad (10)$$

where:

$EF_{best,m,y}$ = Emission factor of power unit m with the best technology commercially available in year y (t CO_2/MWh)

$\eta_{best,y}$ = Power generation efficiency of the best technology commercially available in year y

m = Power units serving the grid with coal, gas, oil or waste incineration in year y

According to the latest and available data at the time of this PD submission, $EF_{grid,BM,y}$ is calculated to be 0.4819 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China[7].

Step 6: Calculate the combined margin (CM) emission factor.

The calculation of the combined margin emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

The weighted average CM method (option A) should be used as the preferred option.

[7] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

The simplified CM method (option b) can only be used if:

a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

b) The data requirements for the application of step 5 above cannot be met.

This PD choose option A.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (11)$$

Where:

$EF_{grid,OM,y}$ = a) operating margin emission factor of NCPG (tCO_2e/MWh)

$EF_{grid,BM,y}$ = b) build margin CO_2 emission factor of NCPG (tCO_2e/MWh)

w_{OM} = c) the weighting of operating margin emission factor (%)

w_{BM} = d) the weighting of build margin emission factor (%)

According to the tool, as a manure treatment project, w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period .

$EF_{grid,CM,,y}$ = 0.9419 * 0.5 + 0.4819 * 0.5 = 0.7119 tCO_2e/MWh

2. Project Emissions

According to the methodology AMS-III.D and AMS-I.D, project emissions consist of:

(a) Physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$);

(b) Emissions from flaring or combustion of the gas stream ($PE_{flare,y}$);

(c) CO_2 emissions from use of fossil fuels or electricity for the operation of all the installed facilities ($PE_{power,y}$);

(d) CO_2 emissions from incremental transportation distances;

(e) Emissions from the storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$).

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \quad (12)$$

Where:

PE_y = Project emissions in year *y* (t CO_2e)

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year *y* (t CO_2e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the biogas stream in the year *y* (t CO_2e)

$PE_{power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year *y* (t CO_2e)

$PE_{transp,y}$ = Emissions from incremental transportation in the year *y* (t CO_2e), as per relevant paragraph in AMS-III.AO

$PE_{storage,y}$ = Emissions from the storage of manure (t CO_2e)

2.1 **Emissions due to physical leakage of biogas in year y**

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect and transport the biogas to the point of flaring or gainful use are estimated as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y} \quad (13)$$

Where:

GWP_{CH4} = Global Warming Potential (GWP) of CH_4 applicable to the crediting period (t CO_2e/t CH_4)

D_{CH4} = CH_4 density (0.00067 t/m^3 at room temperature (20 ºC) and 1 atm pressure)

LT = Index for all types of livestock

j = Index for animal manure management system

$N_{LT,y}$ = Annual average number of animals of type LT in year y (numbers)

$VS_{LT,y}$ = Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)

$MS\%_{i,y}$ = Fraction of manure handled in system i in year y. If the project activity involves sequential manure management systems, the procedure specified in paragraph 18(e) of AMS-III.D shall be used to estimate the project emissions due to physical leakage of biogas in each stage

$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m^3 CH_4/kg-VS), as per paragraph18 of AMS-III.D

2.2 E**missions from flaring or combustion of the biogas stream in the year y**

In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool "Project emissions from flaring" (version 03.0).

Based on the FSR, all the methane generated by the project will be used as energy supply. In order to ensure no biogas is released under exigencies, an emergency flare system is installed at the project site, this emergency flare system is not used under normal operation. In addition, in case this emergency flare system operated, the emissions reductions during this period will be excluded for conservativeness, thus $PE_{flare,y}$ is excluded as well, so $PE_{flare,y}$ = 0 tCO_2e.

2.3 **Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y**

Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool "Project and leakage emissions from anaerobic digesters" (version 02.0), where $PE_{Power,y}$ is the sum of $PE_{EC,y}$ and $PE_{FC,y}$ in the tool.

The project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project. Thus, $PE_{FC,y}$ = 0 tCO_2e.

According to methodological tool "Project and leakage emissions from anaerobic digesters", $PE_{EC,y}$ shall be calculated using the "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0) as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y}) \quad (14)$$

Where:

$PE_{EC,y}$	=	Project emissions from electricity consumption in year y (t CO_2 / yr)
$EC_{PJ,j,y}$	=	Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
$EF_{EL,j,y}$	=	Emission factor for electricity generation for source j in year y (t CO_2/MWh)
$TDL_{j,y}$	=	Average technical transmission and distribution losses for providing electricity to source j in year y
j	=	Sources of electricity consumption in the project

As stated above, according to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation"(version 03.0), for the project is in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the "Tool to calculate the emission factor for an electricity system" (version 07.0) ($EF_{EL,j,y}$ = $EF_{grid,CM,y}$).

As per the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0), the project belongs to the case of Scenario A, use as default values of 20% for project emission, i.e. $TDL_{j,y}$ =20%.

$EC_{PJ,j,y}$ is ex ante determined as 0 in the PD and will be monitored ex post in the verification period.

2.4 **Emissions from incremental transportation in the year y**

According to AMS-III.D, emission from transportation of animal manure between livestock farms and the project site is calculated using relevant paragraph in AMS-III.AO as below:

$PE_{transp,y} = (Q_y/CT_y)*DAF_w*EF_{CO2,transport} + (Q_{res\cdot waste,y}/CT_{res\cdot waste,y})*DAF_{res\cdot waste}*EF_{CO2,transport}$ (15)

Where:

Q_y	=	Quantity of raw waste/manure treated and/or wastewater co-digested in the year y(tonnes)
CT_y	=	Average truck capacity for transportation (tonnes/truck)
DAF_w	=	Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)
$EF_{CO2,\ transport}$	=	CO_2 emission factor from fuel use due to transportation ($kgCO_2$/km, IPCC default values or local values may be used)
$Q_{res\cdot waste,y}$	=	Quantity of residual waste produced in year y (tonnes)
$CT_{res,waste,y}$	=	Average truck capacity for residual waste transportation (tonnes/truck)
$DAF_{res\cdot waste}$	=	Average distance for residual waste transportation (km/truck)

The project site is at the east side of the pasture and waste from the pasture will all enter the receiving tank of the pre-treatment system through tight pipeline, thus there is no emission from transportation of animal manure between livestock farms and the project site. $PE_{transp,y}$ = 0.

2.5 **Emissions from the storage of manure**

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

(a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;

(b) The dry matter content of the manure when removed from the animal barns is less than 20%.

Through satellite maps and on-site visits, the VVB confirms that the breeding farm is very close to the ranch, right next to it. According to the EIA report and actual operation log, the manure from Daming farm is sent to anaerobic digester through pipeline and enter into digester with 24 hours, normally 12 hours, but not exceed 24 hours, and there are no related emissions during the entire project's raw material collection and other processes. hence emissions from the storage of manure is not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

3 . Leakage

As per “Project and leakage emissions from anaerobic digesters” (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

By checking FSR of the project and interviewing with operating staff of the project during site inspection, the VVB confirms that the digestate from the anaerobic digesters of the project will be sent to advanced treatment unit, in mainly involved the filter press as soon as digestate is removed from the digesters and after composting, it will be returned to nearby farm as fertilizer. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

4. **Net GHG emission reductions or removals**

The emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane flared. It is likely that the project activity involves manure treatment steps with higher methane conversion factors (MCF) than the MCF for the manure treatment systems used in the baseline situation, therefore the emission reductions achieved by the project activity are limited to the ex post calculated baseline emissions minus the project emissions using the actual monitored data for the project activity (i.e. N_{LT},y, $MS\%_{i,y}$, as well as $VS_{LT,y}$ in cases where adjusted values for animal weight are used). The emission reductions achieved in any year are the lowest value of the following:

$$\mathrm{ER_{y,ex\ post}} = \min\left[\left(\mathrm{BE_{y,ex\ post}} - \mathrm{PE_{y,ex\ post}}\right), \left(\mathrm{MD_y} - \mathrm{PE_{power,y,ex\ post}}\right)\right] \quad (16)$$

Where:

$ER_{y,ex\ post}$ = Emission reductions achieved by the project activity based on monitored values for year y (tCO_2e)

$BE_{y,ex\ post}$ = Baseline emissions calculated using equation 1. For projects using option in paragraph 17(a) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used

$PE_{y,ex\ post}$ = Project emissions calculated using equation 11 using ex post monitored values of $N_{LT,y,}$ MS% $_{i,y,}$ and if applicable $VS_{LT,y}$

MD_y = Methane captured and destroyed or used gainfully by the project activity in year *y* (tCO_2e)

$PE_{power,y,ex\ post}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year *y* (tCO_2e)

As per para.30 of AMS-III.D, if project activities utilize the recovered methane for power generation, may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \qquad (17)$$

Where:

EG_y = Total electricity generated from the recovered biogas in year *y* (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH4} = NCV of methane (MJ/Nm^3) (use default value: 35.9 MJ/Nm^3)

EE_y = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:

Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation;

Default efficiency of 40 %.

4.1 Calculation of baseline emissions:

Via checking the PD version 1.1/3/ and ER calculation sheet in accordance with the methodology and availbe evidence, CCSC confirmed that the ex-ante value of all parameters to calculate baseline emissions (BEy) and the corresponding source, as listed in the table below, are correct.

Table 3-1: Ex-ante value of parameters to calculate BE_y

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_{2e}/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4}	0.67 kg/m^3	AMS-III.D
UF_b	0.94	AMS-III.D
MCF_j	76%	Table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10
$B_{o,LT}$	Dairy cow:0.13 m^3CH_4/ kg-VS	Table 10.16 of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10
$VS_{LT,y}$	Dairy cow: 1,182.6 kg-dm/ animal/year	$VS_{LT,y}$= VS_{rate} * TAM/1000 * 365 VS_{rate} is 9 kg VS (1000kg animal mass)/day for dairy cow from Asia, which is default value from table 10.13A of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10 TAM is animal mass and 360kg for the project dairy cow. Thus, $VS_{LT,y}$ = 9kgVS(1000kg animal mass)/ day/*360kg/1000 *365 days = 1,182.6kg VS/animal/year
$MS\%_{Bl,j}$	100%	FSR
nd_y	365 days	FSR
$N_{da,y}$	365 days	FSR
$N_{p,y}$	12,000	FSR

As per equations (1), (2), the result of the ex-ante calculated baseline emissions of methane from the manure treatment processes of the project is:

Table 3-2: Ex-ante Baseline Emissions BE_y

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Baseline Emissions (BE_y) tCO_2e

-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{rate} \times \text{TAM}/1000 \text{ X } nd_y$	$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}$
Dairy cow	12,000	1,182.6	24,725
Total	-	-	24,725

As per the FSR, the annual exported electricity to the NCPG of the project is expected to be 16,510 MWh, the combined margin CO_2 emission factor for the grid is 0.7119 tCO_2/MWh, hence the expected annual baseline emissions associated with electricity generation of the project is calculated as follows:

$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$ = 16,510 MWh× 0.7119 tCO_2/MWh = 11,753 tCO_2

$BE_y = BE_{CH4,y} + BE_{elec,y}$ = 24,725 tCO_2e + 11,753 tCO_2e = 36,478 tCO_2e

4.2 **Calculation of project emissions**

Via checking the PD Version 2.0/3/, the ER calculation sheet/4/ as per the applied methodologies and tools of this Project, CCSC confirmed that the ex-ante values of all parameters for project emission calculation are properly selected and consistent with the referred data sources. These parameters and their corresponding sources are summarized in the following table.

Table 3-3: Ex-ante value of parameters to calculate project emissions

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_2e/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4} (ρ_{CH4})	0.67 kg/m^3	AMS-III.D
$B_{o,LT}$	Dairy cow:0.13 m^3CH_4/kg-VS	Table 10.16 of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10
$VS_{LT,y}$	Dairy cow: 1,182.6kg/hd/year	Table 10.13A of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10
$MS\%_{i,y}$	100%	FSR
$N_{da,y}$	365 days	FSR
$N_{p,y}$	12,000	FSR
$Q_{biogas,d}$	8,760,000 m^3/year	FSR

4.2.1 Emissions due to physical leakage of biogas in year y

As described in 4.2, project emissions from physical leakage of biogas are calculated as per equations (2), (12).

Table 4-4: Ex-ante calculate Physical leakage of biogas ($PE_{PL,y}$)

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Physical leakage of biogas ($PE_{PL,y}$) tCO_2e
-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{rate} \times TAM/1000 \text{ X } nd_y$	$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{j,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$
Dairy cow	12,000	1,182.6	3,461
Total	-	-	3,461

So, the ex-ante calculated result of the project's Physical leakage of biogas is 3,461 tCO_2e.

4.2.2 Emissions from flaring or combustion of the biogas stream in the year y

As described above, $PE_{flare,y}$ is excluded, hence $PE_{flare,y}$ = 0 for the project.

4.2.3 Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

As described in 4.2, the project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project emission.

As described in 4.2, the electricity consumption of the project site can be met by the generated electricity. Thus, $PE_{EC,y}$ = 0.

4.2.4 Emissions from incremental transportation in the year y

As described above, $PE_{transp,y}$ is excluded, hence $PE_{transp,y}$ = 0 for the project.

4.2.5 Emissions from the storage of manure

As described above, $PE_{storage,y}$ is excluded, hence emissions from the storage of manure are not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

So, the ex-ante estimated project emission shall calculate as per equation (10):

$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} = PE_{PL,y} = 3{,}461 tCO_2e$

4.3 Calculation of leakage

As per "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Digestate from the anaerobic digesters of the project will be used to produce fertilizer as soon as digestate is removed from the digesters. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

Thus, $LE_y = 0$.

4.4 C**alculation of emission reductions (Ex ante calculation)**

Estimated annual emission reductions of the project can be calculated as following equation (19):

$$ER_y = BE_y - PE_y - LE_y = BE_y - PE_y \quad (19)$$

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
26-August-2022~31-December-2022	12,792	1,214	0	11,579
01-January-2023~31-December -2023	36,478	3,461	0	33,017
01-January -2024~31-December -2024	36,478	3,461	0	33,017
01-January -2025~31-December -2025	36,478	3,461	0	33,017
01-January -2026~31-December -2026	36,478	3,461	0	33,017
01-January -2027~31-December -2027	36,478	3,461	0	33,017
01-January -2028~31-December -2028	36,478	3,461	0	33,017
01-January -2029~25-August -2029	23,686	2,247	0	21,438
Total	**255,346**	**24,227**	**0**	**231,119**

In conclusion, the assessment team confirms the following:

- All assumptions and data used by the project participants are listed in the PD (version 2.0 dated 17-February-2024) and/or supporting documents, including their references and sources;

- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PD (version 2.0 dated 17-February-2024);

- All values used in PD(version 2.0 dated 17-February-2024) are considered reasonable in the context of the project activity;

- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;

- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in PD (version 2.0 dated 17-February-2024).

CCSC confirms that the methodologies and the above referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals.

3.3.7 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV's request for deviation of Chinese project activities for the CDM baseline methodology AM0005 has been applied for calculating the build margin (BM) emission factor of this project activity. Moreover, the result of BM and OM are sourced from local government.

CCSC confirms that the deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals.

3.3.8 Monitoring Plan

1. Data and parameters available at validation

Each parameter determined ex-ante only used to calculate the ex-ante emission reduction. These parameters and corresponding assessment are listed in the table below.

GWP_{CH4}	**Global Warming Potential (GWP) of CH_4 applicable to the crediting period**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, tCO_2e/tCH_4

Appropriate description?	Yes
Source clearly referenced?	The source is default value from IPCC Fifth Assessment Report (AR5) as per the applied methodology AMS-III. D.(Version 21.0)
Correct value provided?	Yes. The applied value is 28/tCO_2e/tCH_4
Has this value been verified?	The value is confirmed via checking the IPCC AR5.
Choice of data correctly justified?	Yes

D_{CH4}	CH_4 density
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, kg/m^3
Appropriate description?	Yes
Source clearly referenced?	The data source is para. 18 of the applied methodology AMS-III. D.(Version 21.0)
Correct value provided?	Yes. The applied value is 0.67kg/m^3
Has this value been verified?	The value has been verified via checking the methodology AMS-III.D.(version21.0)
Choice of data correctly justified?	Yes

UF_b	Model correction factor to account for model uncertainties
Title in line with Methodology?	Yes
Data unit correctly expressed?	N/A
Appropriate description?	Yes

Source clearly referenced?	The data source is para. 18 of the applied methodology AMS-III. D.(Version 21.0)
Correct value provided?	Yes. The applied value is 0.94, which is consistent with AMS-III.D(version 21.0).
Has this value been verified?	The value has been verified via checking the methodology AMS-III.D.(version21.0)
Choice of data correctly justified?	N/A

MCF_j	**Annual methane conversion factor (MCF) for the baseline animal manure management system j**
Title in line with Methodology?	Yes
Data unit correctly expressed?	N/A
Appropriate description?	Yes
Source clearly referenced?	The data source is table 10.17of 2019 Refinement to the 2006 IPCC guidelines as per the AMS-III.D.(version 21.0)
Correct value provided?	Yes. The applied value is 76%.
Has this value been verified?	The value has been verified through checking the methodology AMS-III.D.(version21.0)
Choice of data correctly justified?	Yes

$B_{O,LT}$	**Maximum methane producing potential of the volatile solid generated for animal type LT**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, m^3 CH_4/kg-VS

Appropriate description?	Yes
Source clearly referenced?	The data source is table 10.16 of 2019 Refinement to the 2006 IPCC guidelines.
Correct value provided?	Yes. The applied value is 0.13 m^3 CH_4/kg-VS.
Has this value been verified?	The value has been verified through checking the 2019 Refinement to the 2006 IPCC guidelines. The project adopts Asian low productivity data.
Choice of data correctly justified?	Yes

$MS\%_{BL,i}$	Fraction of manure handled in baseline animal manure management system j
Title in line with Methodology?	Yes
Data unit correctly expressed?	N/A
Appropriate description?	Yes
Source clearly referenced?	The data source is FSR
Correct value provided?	Yes. The applied value is 100%
Has this value been verified?	The value has been verified through checking the FSR
Choice of data correctly justified?	N/A

$N_{da,y}$	Number of days animal is alive in the farm in the year y
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, Day
Appropriate description?	Yes

Source clearly referenced?	The source of data is the statement of the project proponent.
Correct value provided?	Yes. The applied value is 365.
Has this value been verified?	The value has been verified via interview with PP.
Choice of data correctly justified?	Yes

$EF_{grid,OM,y}$	**Operating margin CO_2 emission factor in year y**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, t CO_2/MWh
Appropriate description?	Yes
Source clearly referenced?	The source of data is the "2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Correct value provided?	Yes. The applied value is 0.9419
Has this value been verified?	The value has been verified via checking the "2019 Baseline Emission Factors for Regional Power Grids in China"/28/. The value for NCPG is 0.9419 t CO_2/MWh.
Choice of data correctly justified?	Yes

$EF_{grid,BM,y}$	**Build margin CO_2 emission factor in year y**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, t CO_2/MWh
Appropriate description?	Yes
Source clearly referenced?	The source of data is the "2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China

Correct value provided?	Yes. The applied value is 0.4819
Has this value been verified?	The value has been verified via checking the "2019 Baseline Emission Factors for Regional Power Grids in China"/28/. The value for NCPG is 0.4819 t CO_2/MWh.
Choice of data correctly justified?	Yes

$EF_{grid,CM,y}$ ($EF_{grid,y}$, $EF_{EL,j,y}$)	**Combined margin CO_2 emission factor in year y**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, t CO_2/MWh
Appropriate description?	Yes
Source clearly referenced?	Calculated as per the "Tool to calculate the emission factor for an electricity system" (version 07.0).
Correct value provided?	Yes. The applied value is 0.7119
Has this value been verified?	The value indicated in PD and ER spread sheet has been verified via calculation based on $EF_{grid,OM,y}$ and $EF_{grid,BM}$ and confirmed as correct.
Choice of data correctly justified?	Yes

$TDL_{j,y}$	**Average technical transmission and distribution losses for providing electricity to source j in year y**
Title in line with Methodology?	Yes
Data unit correctly expressed?	N/A
Appropriate description?	Yes
Source clearly referenced?	Default value from " Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version

	03.0)
Correct value provided?	Yes. The applied value is 20%.
Has this value been verified?	The value has been verified through checking "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0).
Choice of data correctly justified?	Yes

NCV_{CH4}	**NCV of methane**
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes, MJ/Nm^3
Appropriate description?	Yes
Source clearly referenced?	Default value from AMS-III.D
Correct value provided?	Yes. The applied value is 35.9 MJ/Nm^3
Has this value been verified?	The value has been verified through checking para 30 of AMS-III.D.
Choice of data correctly justified?	Yes

2、Data and parameters monitored

The monitoring parameters required by the methodology and applicable tools for the project are summarized in the below table.

$MS\%_{i,y}$	**Fraction of manure handled in system i in year y**
Title in line with Methodology?	Yes
Ex-ante value	100%
Appropriate description?	Yes
Source clearly referenced?	The ex-ante value source is FSR.

Correct value provided?	Yes. The applied value is 100%. It has been verified via checking the FSR.
Choice of data correctly justified?	Yes
Measurement method correctly described?	Yes. All manure would be handled in this project.
Monitoring frequency correctly described?	Annually, based on daily measurement and monthly aggregation, which is confirmed as consistent with methodology AMS-III.D.
Monitoring equipment correctly described?	N/A
QA/QC procedures	N/A
Purpose of data	Calculation of baseline emissions and project emissions

$N_{p,y}$	**Number of animals produced annually of type LT for the year y**
Title in line with Methodology?	Yes
Ex-ante value	12000
Appropriate description?	Yes
Source clearly referenced?	The ex-ante value source is FSR. The applied value in PD and ERs spread sheet is verified to be consistent with that of FSR.
Correct value provided?	Yes. The applied value is 12000
Choice of data correctly justified?	Yes
Measurement method correctly described?	Yes. The number of Dairy cows produced in the farm will be recorded manually by the responsible staff.
Monitoring frequency correctly described?	Annually, based on monthly records
Monitoring equipment correctly described?	N/A

QA/QC procedures	N/A
Purpose of data	To calculate the annual average number of animals ($N_{LT,y}$)

nd_y	**The number of days the treatment plant was operational in year y.**
Title in line with Methodology?	Yes
Ex-ante value	365
Appropriate description?	Yes
Source clearly referenced?	The Ex-ante value is from FSR.
Correct value provided?	Yes. The applied value is 365
Choice of data correctly justified?	Yes
Measurement method correctly described?	Yes. 365 days used for ex-ante estimation. The actual number of days that the treatment plant is in operation during the monitoring periods will be monitored and recorded by operational staff.
Monitoring frequency correctly described?	Annually, based on daily records and monthly aggregation
Monitoring equipment correctly described?	N/A
QA/QC procedures	N/A
Purpose of data	To estimate the annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis (VSLT,y).

$VS_{LT,y}$	**Volatile solids for livestock LT entering the animal manure management system in year y**
Title in line with Methodology?	Yes

Ex-ante value	1,182.60
Appropriate description?	Yes
Source clearly referenced?	The Ex-ante value is resulted from the calculation using the default value from table 10.13A of 2019 Refinement to the 2006 IPCC Guidelines based on the equation 10.22 A of the same document. VS is calculated according to equation as follows： $VS_{LT,y} = VS_{rate} * TAM/1000 * 365$ VS_{rate} is the default value 9 kg VS (1000kg animal mass)/day for dairy cow from Asia. TAM is animal mass and 360kg for the project dairy cow. The weight of dairy cow 360kg was used in ER ex-ante estimation, during verification period, actual average animal mass will be used to adjust the $VS_{LT,y}$ and ER ex-post calculation.
Correct value provided?	Yes. The applied value is 1,182.60 kg-dm/animal/year
Choice of data correctly justified?	Yes
Measurement method correctly described?	Yes
Monitoring frequency correctly described?	Annually, based on daily records and monthly aggregation
Monitoring equipment correctly described?	N/A
QA/QC procedures	N/A
Purpose of data	Calculation of baseline emissions and project emissions

$EG_{PJ,y}$	**Quantity of electricity generation supplied by the project to the grid in year y**

Title in line with Methodology?	Yes
Ex-ante value	16,510 MWh
Appropriate description?	Yes
Source clearly referenced?	The Ex-ante value is from FSR
Correct value provided?	Yes. The applied value has been checked with FSR.
Choice of data correctly justified?	Yes
Measurement method correctly described?	The actual data in the monitoring period will be continuously measured by electricity meter and recorded at least monthly.
Monitoring frequency correctly described?	Yes. The data will be continuously monitored, hourly measured and at least monthly recorded.
Monitoring equipment correctly described?	Electricity meter (M1) Type: DSZ178 Serial number: 4110013023 Accuracy: 0.2S
QA/QC procedures	The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation in accordance with "DL/T448 – 2016 Technical Administrative Code of Electric Energy Metering". The accuracy of the meter is not less than 0.5s. The monitoring data will be cross-checked with ETN issued by the Grid.
Purpose of data	Calculation of baseline emissions

EG_y	**Total electricity generated from the recovered biogas in year y**
Title in line with Methodology?	Yes
Ex-ante value	16,510 MWh
Appropriate description?	Yes
Source clearly referenced?	The Ex-ante value is from FSR
Correct value provided?	Yes. The applied value has been checked with

	FSR.
Choice of data correctly justified?	Yes
Measurement method correctly described?	The actual data in the monitoring period will be continuously measured by electricity meter and recorded. EG_y is the total electricity generated from the recovered biogas in year y. $EG_{PJ,y}$ is the quantity of electricity generation supplied by the project to the grid in year y. Using $EG_{PJ,y}$ to replace EG_y is more conservative. In addition, no matter it is EG_y or $EG_{PJ,y}$, it does not influence the amount of the final emission reduction since after the minimization according to the methodology of $\left(BE_{y,ex\ post} - PE_{y,ex\ post}\right)$ and $\left(MD_y - PE_{power,y,ex\ post}\right)$, the value of $\left(BE_{y,ex\ post} - PE_{y,ex\ post}\right)$ is taken.
Monitoring frequency correctly described?	Yes. The data will be continuously monitored, hourly measured and at least monthly recorded.
Monitoring equipment correctly described?	Electricity meter (M1) Type: DSZ178 Serial number: 4110013023 Accuracy: 0.2S
QA/QC procedures	The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation. The accuracy of the meter is not less than 0.5s. The monitoring data will be cross-checked with ETN issued by the Grid.
Purpose of data	Calculation of emission reductions

$EC_{PJ,j,y}$	**Quantity of electricity consumed by the project in year y**
Title in line with Methodology?	Yes
Ex-ante value	0
Appropriate description?	Yes
Source clearly referenced?	The Ex-ante value is determined as 0 in the PD

	according to the FSR.
Correct value provided?	Yes. The applied value has been checked.
Choice of data correctly justified?	Yes
Measurement method correctly described?	The actual data in the monitoring period will be continuously measured by electricity meter and recorded.
Monitoring frequency correctly described?	Yes. The data will be continuously monitored and monthly recorded.
Monitoring equipment correctly described?	Electricity meter (M1) Type: DSZ178 Serial number: 4110013023 Accuracy: 0.2S
QA/QC procedures	Purchase receipts are to be used for crosscheck. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation. The accuracy of the meter is not less than 0.5s.
Purpose of data	Calculation of project emissions

EE_y	**Energy Conversion Efficiency of the project equipment**
Title in line with Methodology?	Yes
Ex-ante value	/
Appropriate description?	Yes
Source clearly referenced?	The default value is from ASM-III. D
Correct value provided?	/
Choice of data correctly justified?	Yes
Measurement method correctly described?	As per paragraph 30 Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for this fuel. If the specification provides a range of efficiency values,

	the highest value of the range shall be used for the calculation.
Monitoring frequency correctly described?	/
Monitoring equipment correctly described?	/
QA/QC procedures	/
Purpose of data	Calculation of emission reductions

In conclusion, CCSC confirms that:

- The data and parameters fixed ex-ante have been correctly listed. Parameters fixed ex-ante for required parameters have been verified by checking the information flow and in compliance with the monitoring plan of the PD version 2.0/3/.
- All parameters required by the monitoring plan have been correctly listed. The monitored data for required parameters have been verified by checking the whole information flow and in compliance with the monitoring plan of the PD version 2.0/3/.

3. Monitoring plan

The approved baseline and monitoring methodology AMS-III.D (ver. 21.0) and AMS-I.D (ver. 18.0), have been applied. The monitoring diagram of the project is shown in Figure 3-1.

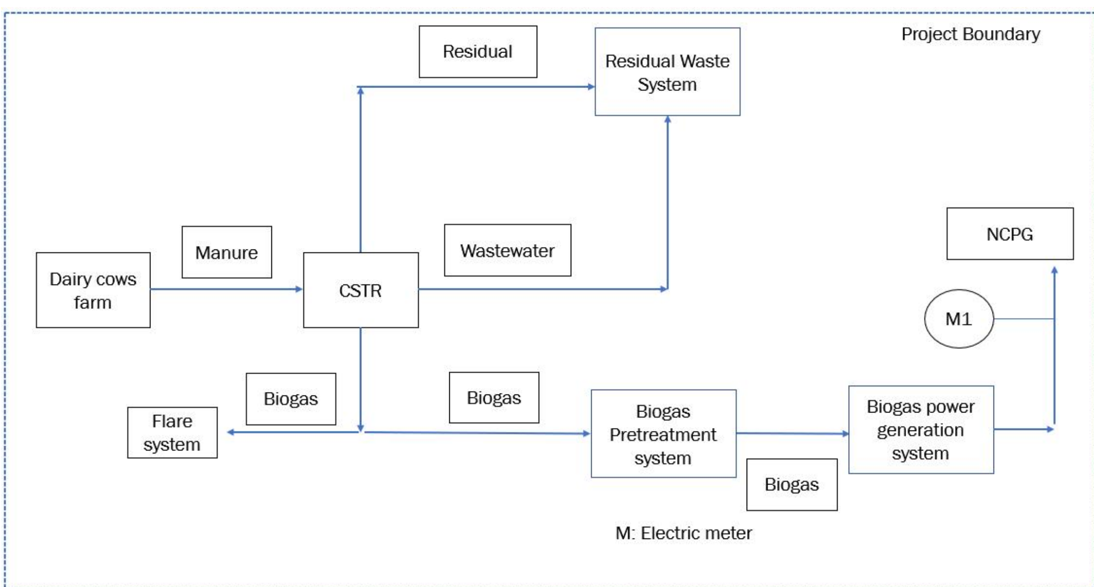


Figure 3-1 Diagram of the project

The validation team has checked the monitoring plan described in the PD against the applied methodology. The monitoring plan in the PD version 2.0/3/has been designed to comply with the latest applicable version of the methodology (AMS-III.D Ver.21.0 and AMS-I.D.)/6//7/.

The validation team evaluated the feasibility and sufficiency of the monitoring plan. The key components of the monitoring plan are as follows.

（A） Monitoring equipment and Installation

Installation and configuration of meters are shown in a diagram in the PD /1//3/. To ensure measurements with a low degree of uncertainty, the data metering equipment will be calibrated and checked by an appropriately qualified third party according to an appropriate national standard. The calibration records will be appropriately maintained and made available for review by VVB. Figure 3-1 below shows the diagram of project monitoring system.

（B） Monitoring Structure

The PD /1//3/ contains a flowchart illustrating the Organization Structure of the Monitoring Team responsible for data collection, supervision and witness the whole process of data measuring and recording. A VCS manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement are to be carried out by designated monitoring officers. In addition, the project developer appoints internal verifiers who are responsible for internal check of the measurement, collection of relevant receipts and invoices, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties is developed and served as the basis of the project monitoring. The Figure 3-2 below shows the operation and management structure of the Project.

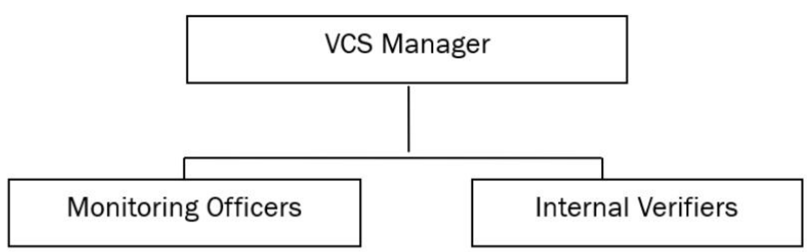


Figure 3-2. Operation and management structure of the project

（C） Data collection

Monitoring officers are responsible for data collection. Designated teams will read and collect the monitored data regularly. The computer system will automatically monitor and record relevant meter data. Automatic records will serve as the main data source for emission reductions calculation. All data files, relevant flow rate records will be collected by a designated monitoring officer, who will prepare backup in time and archive all documents properly.

（D） Quality assurance

All metering equipment for monitoring will be chosen in accordance with VCS requirements/12//13//14/, and will be calibrated regularly for accuracy by qualified party according to the national regulations. To assist in future verification, the Project owner will preserve the calibration records, along with the data files of project monitoring as critical evidences.

Error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively. All the meters will be checked and maintained periodically.

Sales and purchase receipts are to be used for crosscheck. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out once a year to ensure normal operation. The accuracy of the meter is not less than 0.5s.

（E） Data file management

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy. Other documents in paper, for instance. maps, forms and environment assessment reports will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period. The Project owner will provide original records and documents if necessary.

Via checking the monitoring plan in the PD version 2.0/3/as described above, examining the relevant records and equipments, interviewing with operating staff of the project during site inspection, CCSC validation team confirms that the monitoring plan contains all necessary parameters which have been clearly described in PD and that the means of monitoring described in the monitoring plan complies with the requirements of the methodology. A Management Structure of the Monitoring Team is provided in the PD. The functions such as data collection, aggregation, verification, calculation, archiving, as well as the maintenance of equipment etc. have been defined. Quality assurance and quality control procedures for recording, maintaining and data archiving etc. will be ensured according to Verra’s rules. The monitoring system equipment will be implementation properly as per relevant standard and regulations. Via checking the meter nameplate, the accuracy of the meter is 0.2s， not less than 0.5s. The monitoring data would be cross checked via sales receipts for the purpose of quality control. The project owner will record the readings of the meter monthly. The calibration of the meters will be implemented once a year as per relevant national standards to ensure normal operation. An emergency treatment process has also been addressed in the PD version 2.0 when the meter is in malfunction.

In conclusion, based on document review, on-site inspection and stakeholder interview, together based on CCSC's local and sectoral expertise, CCSC confirms that:

-The monitoring plan is in compliance with the requirements of the methodology. The monitoring plan developed by the PP comprehensively covers all relevant data and parameters and follows guidance in the PD template, and the information contained therein has been validated.

-Monitoring arrangements described in the monitoring plan are feasible within the project design.

-The PP's ability to implement the monitoring plan can be guaranteed.

3.4 Non-Permanence Risk Analysis

This section is not applicable for the current Project as it is not a AFOLU project.

4 VALIDATION OPINION

CCSC has been engaged by Climate Bridge (Shanghai) Ltd. to perform the validation of Hebei Handan Damingchang Biogas Recovery and Utilization Project located in Hebei Province, P.R.China.

The management of the project proponent is responsible for the preparation of the GHG emissions data and the estimated GHG emissions reductions on the basis set out within the project's Monitoring Plan in the PD and the approved methodology AMS-III.D (version21.0). and AMS-I.D.(Version 18.0).

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VERRA. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these.

A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided CCSC with sufficient evidence for positive validation opinion as per the requirement of VCS Standard Version 4.4.

Through documented evidence reviewing and by an on-site assessment, CCSC can confirm that:

➢ the baseline scenario is correctly defined as per the applied methodology and relate tools;

➢ the project is additional;

➢ all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;

➢ All data and parameters along with their references and sources used for determine the ex-ante of emission reduction, are listed, and correctly quoted and in the PD (version 2.0)/3/ ,

➢ All values used for calculating project emission reductions are considered reasonable in the context of the project activity.

➢ the monitoring plan in the validated PD/3/ is as per the applied baseline and monitoring methodology.

➢ the project has been implemented as per the VCS PD;

➢ the project complies with the validation criteria for projects set out in VCS Standard

Version 4.4;

➢ the project description and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS requirement and methodologies AMS-III.D.(Version 21.0) and AMS-I.D.(version 18.0);

➢ the monitoring mechanism is in place as per the applied baseline and monitoring methodology;

The estimated GHG emission reductions and removals in the first 7-year crediting period (from 26-August-2022 to 25-August-2029) is validated as the following:

Validated GHG emission reductions and removals in the above period:

Year	Estimated GHG emission reductions or removals (tCO_2e)
26-August-2022~31-December-2022	11,579[8]
01-January-2023~31-December -2023	33,017
01-January -2024~31-December -2024	33,017
01-January -2025~31-December -2025	33,017
01-January -2026~31-December -2026	33,017
01-January -2027~31-December -2027	33,017
01-January -2028~31-December -2028	33,017
01-January -2029~25-August -2029	21,438[9]
Total estimated ERs	231,119
Total number of crediting years	7
Average annual ERs	33,017

[8] It covered 128 days from 26-August-2022~31-December-2022, thus the estimated emission reduction during this period is 11,579 tCO_2e = 33,017 tCO_2e x 128/365.

[9] It covered 237 days from 01-January-2029~25-August-2029, thus the estimated emission reduction during this period is 21,438 tCO_2e = 33,017 tCO_2e x 237/365.

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCSC	China Classification Society Certification Co. Ltd.
NCPG	North China Power Grid
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CL	Clarification request
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
DOE	Designated Operational Entity
DNA	Designated National Authority
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
FAR	Forward Action Request
GHG	Greenhouse Gas(es)
PD	Project Description
PP	Project Proponent
UNFCCC	United Nations Framework Convention on Climate Change
VVB	Validation/Verification Body
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Ref no.	Reference Document
/1/	Climate Bridge (Shanghai) Ltd. VCS Project Description, Version 1.0 dated 08-November-2022, Version 1.1 dated 25-March-2023
/2/	Climate Bridge (Shanghai) Ltd. ER calculation spreadsheet, Version 01 dated 08-November-2022, Version 2.0 dated 17-February-2024
/3/	Climate Bridge (Shanghai) Ltd. VCS Project Description, Version 2.0 dated 17-February-2024
/4/	Climate Bridge (Shanghai) Ltd. Project LCOH calculation spreadsheet Version 02 dated 25-March-2023
/5/	VERRA, VCS standard v4.4，17-January-2023
/6/	CDM Executive Board, AMS-III.D. Methane recovery in animal manure management systems (Version 21.0) dated 22-September-2017
/7/	CDM Executive Board, AMS-I.D. Grid connected renewable electricity generation (Version 18.0) dated 28-Novermber-2014
/8/	CDM Executive Board, Tool to calculate the emission factor for an electricitysystem, (version 07.0), 31-August-2018
/9/	CDM Executive Board, Project and leakage emissions from anaerobic digesters (version 02.0), 22-September-2017
/10/	CDM Executive Board, Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0), 22-September-2017
/11/	CDM Executive Board, Project emissions from flaring (version 4.0), 11-March-2022
/12/	VERRA, VCS Program Guide Version 4.3 dated 17-January-2023

/13/	VERRA, Registration & Issuance Process Version 4.3,17-January-2023
/14/	VERRA, Methodology-Requirements Version 4.3, 17-January-2023
/15/	Business license of Daming Jiujin Energy Co., issued on 02-November-2020
/16/	Administrative Approval Bureau of Daming County, EIA approval of the Project dated 9-September-2021.
/17/	Administrative Approval Bureau of Daming County, Project approval dated 23-April-2021
/18/	Construction document dated 06-Febuary-2022
/19/	Equipment purchase contracts
/20/	Hebei Power Grid Company, Approval for grid-connection, approved on 24-August-2022
/21/	Power Purchase Agreement, signed between the PP and local Power Grid Company
/22/	Acceptance report of test operation of generator set, 29-September-2022
/23/	Environmental Impact Assessment Report, June-2021
/24/	Feasibility study report, April-2021
/25/	EIA report preparer qualification certificate
/26/	IPCC 19R_V4_Ch10, Emissions from livestock and manure management
/27/	Questionnaire survey dated 08-July-2021
/28/	Chinese DNA, 2019 Baseline Emission Factors for Regional Power Grids in China dated 29-December-2020

/29/	Qualification certificate of FSR preparation agency
/30/	VERRA, VCS validation report template Version 4.2, 21-December-2022
/31/	Dynamic table of Damingchang farm
/32/	Project Acceptance Report dated 28-January-2022
/33/	Operation log
/34/	Nameplates of the main equipment and meters
/35/	VERRA, Vcs-project-description-template-v4.2, 21-December-2022
/36/	VERRA, VCS program definitions Version 4.3, 21-December-2022

APPENDIX C: RESOLUTION OF CLARIFICATION REQUEST AND CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

Table 1. CL from this verification

CL ID	CL1	**Section no.**	2.2	**Date:**
Description of CL				
The original survey questionnaires collected from local residents were not included in the evidence documentation provided by PP.				
Project participant response				**Date:** 21/03/2023
The original survey questionnaires has been provided.				
Documentation provided by project participant				
The original survey questionnaires				
DOE assessment				**Date:** N/A
The PP submitted the original survey questionnaires to the validation team. Through reviewing the document, the validation team confirmed that the evidence was correct and adequate to prove . Thus CL-1 was closed successfully.				

Table 2. CAR from this verification

CAR ID	CAR1	**Section no.**	1.1	**Date:**20-March-2023
Description of CAR				
The template used in the PD Version 1.0 doesn’t conform with the latest version of the PD template, V4.2				
Project participant response				**Date:** 25-March-2023
PD template V4.2 has been used.				
Documentation provided by project participant				
PD-V2.0				

DOE assessment	**Date:** 25-March-2023
The PP resubmitted the updated the PD-V2.0. The assessment team confirmed that the updated PD-V2.0 is in accordance with the latest PD template V4.2. Therefore, CAR 1 was successfully closed.	

CAR ID	CAR2	**Section no.**	1.3	**Date:**20-March-2023
Description of CAR				
The VCS standard applied in the PD Version 1.0 is not the latest available version 4.4, which is effective from 21-december-2022.				
Project participant response				**Date:** 25-March-2023
VCS standard V4.4 has been used.				
Documentation provided by project participant				
PD-V2.0				
DOE assessment				**Date:** 25-March-2023
The PP resubmitted the updated the PD-V2.0. Through reviewing the resubmitted PD-V2.0, the validation team confirmed that the updated PD-V2.0 has been modified based on the VCS standard V 4.4. Hence, CAR 2 was successfully closed.				

Table 3. FAR from this verificatio

FAR ID	N/A	**Section no.**	N/A	**Date:** N/A
Description of FAR				
N/A				
Project participant response				**Date:** N/A
N/A				
Documentation provided by project participant				
N/A				
DOE assessment				**Date:** N/A
N/A				